

pd-trigger — USB-C Power Delivery Trigger Board

Design Document, rev. v1 (S14 run c)

ai-ee pipeline — AI-generated design, human-reviewed gates

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Abstract

A bench USB-C Power Delivery *sink* trigger: negotiates a user-selected PD profile (5/9/12/15/20 V) from any PD source and delivers it on a screw terminal, sized for 5 A continuous (100 W at 20 V). CH224K sink controller, DIP-switch profile straps, zener-window fallback indication, 48×30 mm, 2-layer **2 oz** copper, JLCPCB economy assembly. All five machine gates green (ERC, placement, routed DRC, 8-check verification, gerber DFM); two adversarial design reviews passed. Estimated \$19.65 for 10 assembled (quote-page price is authoritative).

1 Requirements

- USB-C receptacle, PD **sink** only — no USB data. “PD SRC ONLY” is printed at the connector: legacy QC/BC adapters supply 5 V fallback only (by design, see §3).
- Output profiles 5/9/12/15/20 V, selected on-board; uniform **5 A** rating at every profile; screw terminal (≥ 8 A, 18 AWG) plus a low-current auxiliary header (“AUX 1A MAX” silk).
- Fallback to 5 V when the source cannot deliver the selected profile is acceptable but must be *visibly indicated*.
- Bench use, no enclosure, no battery; prototype qty 10; JLC Basic parts preferred; soft size target $\sim 40 \times 25$ mm (+20% allowance used: final 48×30 mm).

2 Architecture

Signal/power chain:

J1 USB-C → VBUS pour fan-in → TVS D1 + bulk C1A/C1B/C2 → **3.0 mm trunk** → J2 screw terminal
taps: R14 0R → /VIND housekeeping; F1 PPTC 1A → /VAUX → J3 aux
/VIND → dropper R2A+R2B ($2 \times 510 \Omega$, 1206) → /VDD (CH224K shunt node, C5 1 μ F)

2.1 Key components

Role	Part	Note
PD sink controller	WCH CH224K (ESSOP-10)	CFG straps, no MCU
ESD/surge	TI TVS2200 (flat clamp)	22 V standoff / 28.4 V clamp
Receptacle	GCT USB4105-GF-A-120	5 A collective, 48 V rated
Output	KF128-class 5.08 mm terminal	≥ 8 A
Selector	3-pos DIP to GND	open = 5 V (fail-safe)
Indication	zener 6.2 V + BC847BS window	red “5V” / green “OK”

3 Design decisions of record

1. **No series protection in the 5 A path.** The original requirement asked for a PTC; research showed no SMD PPTC holds 5 A at 40 °C and a compliant PD source folds back at ~5–6 A — below any such part’s trip point. Protection chain: source OCP (per PD spec) + mandatory TVS + 3×-derated copper. The resettable element moved to the 1 A aux tap where its trip window is real.
2. **Dropper, not LDO, for the controller.** The CH224K’s VDD pin is an internal *shunt* regulator (3.24–3.36 V, sinks 0–30 mA); the datasheet’s own reference circuit feeds it via a 1 kΩ dropper. A 3.3 V LDO output can exceed the shunt band and fight it. Implemented as 2×510 Ω in series (1206 halves, 0.16 W each at 21 V, 3× derated). A prior field failure attributed to this topology was a *sizing* error (510 Ω total in a small package).
3. **Strap budget.** At the 5 V profile the dropper delivers only ~1.7 mA, so CFG pull-ups are 100 kΩ to /VDD (99 μA worst case); worst-corner load computed at 0.979 mA. CH224K quiescent current is unpublished — *bring-up step: measure /VDD at the 5 V profile* (contingency: reduce dropper toward 600 Ω total).
4. **DP/DM shorted at the chip** (PD-only mode), connector data pins unconnected. Reviewer-ruled *safer* than the datasheet’s connector wiring (removes a 21 V-onto-3.8 V-abs-max fault path); the cost is no legacy QC/BC negotiation — hence “PD SRC ONLY”.
5. **Fail-safe profile straps.** DIP switches short CFG lines to GND against 100 k pull-ups; an open/broken switch selects $\text{CFG1} = 1 \Rightarrow 5 \text{ V}$.
6. **2 oz outer copper.** IPC-2152 at 5 A, $\Delta T = 10^\circ\text{C}$: 1.75 mm at 2 oz vs. an unroutable 3.5 mm at 1 oz. *Must be selected on the JLC order page.*

4 Profile selection and indication

4.1 Switch table (printed on the back silk)

ON = strap to GND (logic 0). The raw CFG bit table is *inverted* relative to switch positions — the printed table is the authority:

Profile	SW-1 (CFG1)	SW-2 (CFG2)	SW-3 (CFG3)
5 V	OFF	any	any
9 V	ON	ON	ON
12 V	ON	ON	OFF
15 V	ON	OFF	OFF
20 V	ON	OFF	ON

4.2 Indication (no MCU)

A 6.2 V zener + dual-NPN window on VBUS trips at 6.15–7.25 V across corners (clears 5.25 V by $\geq 0.9 \text{ V}$, 8.55 V by $\geq 1.3 \text{ V}$) and drives mutually-exclusive LEDs: red **5V** = source refused the profile (5 V fallback active), green **OK** = negotiated profile $\geq 9 \text{ V}$ delivered. **PWR** always-on from /VIND. Verified 1.3–3.8 mA LED currents at all five profiles; the window transistor’s off-state collector sees 21 V against a 45 V V_{CEO} .

5 PCB implementation

- Stackup JLC2313.1.6_2oz: 2 layers, 1.6 mm, 70 μm outer copper. Outline 48×30 mm.

- **5 A forward path:** J1's paired VBUS contacts merge in an F.Cu pour (3.5 mm effective copper — rule-width tracks cannot reach the pads past the CC pins, measured 1.465 mm max), then a single 3.0×32 mm F.Cu trunk to J2. Zero vias in the VBUS path; computed rise $\sim 4^\circ\text{C}$ at 5 A.
- **5 A return:** B.Cu ground pour, unslotted beneath the trunk. J1's ground contacts tie into priority-1 F.Cu pour lobes (solid connect) spanning the SMD contacts and THT shell pegs, with **16 ground vias** at the connector: per-pad attachment 2.09 mm / **5.68 A** and 1.62 mm / **4.70 A** against a ~ 2.5 A per-pad requirement.
- TVS stub < 10 nH loop (3.6 mm from J1, 3-via return); bulk capacitance $20\ \mu\text{F}$ at the connector ($< 100\ \mu\text{F}$ PD ceiling, $10\ \mu\text{F}$ inrush budget honored).
- Functional silk: “PD SRC ONLY”, terminal $+/-$, “AUX 1A MAX”, pin-locked LED legends, “PROFILE SEE BACK” pointing at the mirrored B-side table.

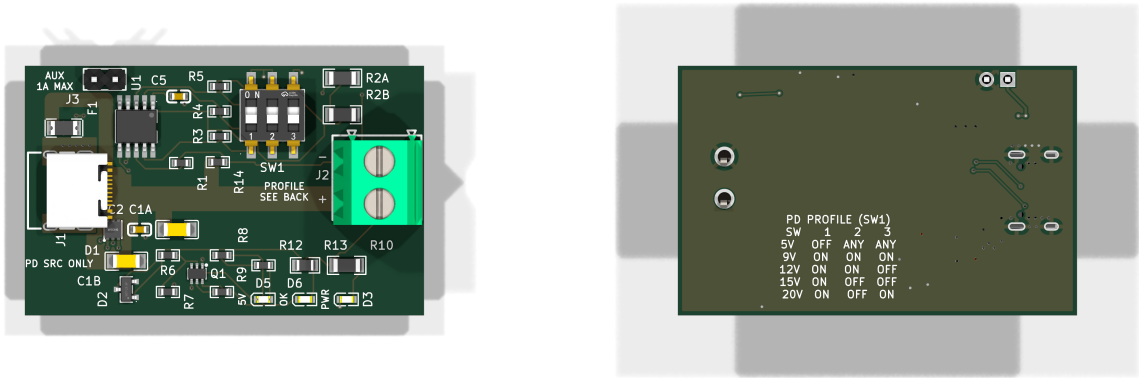


Figure 1: Assembled-view renders. Left: top — USB-C at left, 5 A corridor to the screw terminal at right, DIP selector top edge, LED row bottom. Right: bottom — ground pour and the mirrored profile table.

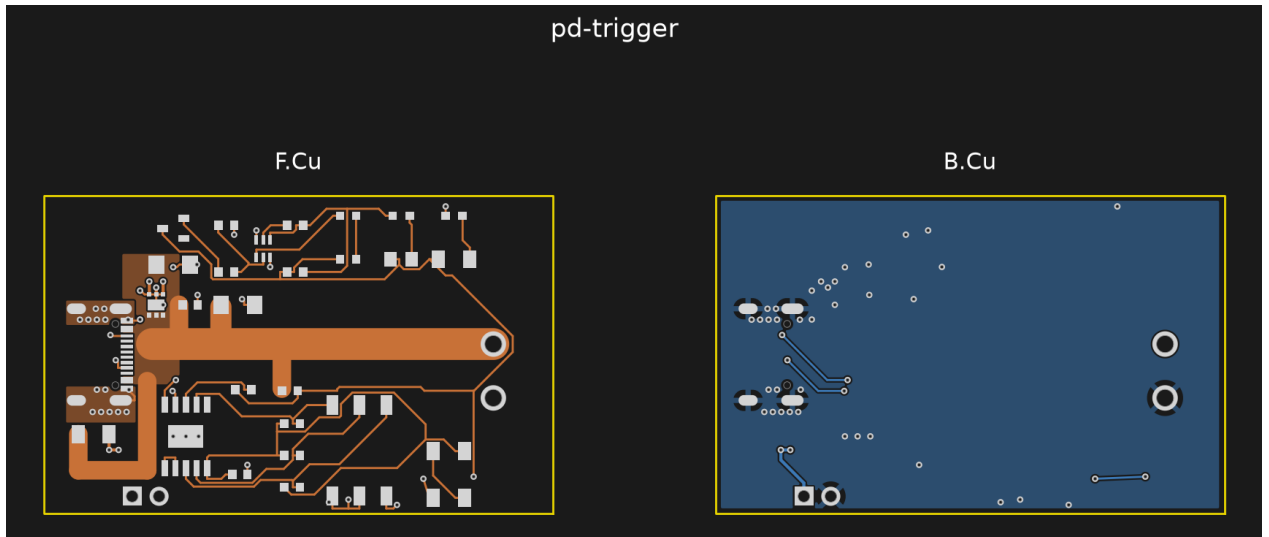


Figure 2: Copper layers from the fabrication gerbers. F.Cu: pour fan-in at J1, 3 mm trunk, ground lobes. B.Cu: return pour, ground stitching.

6 Verification record

Gate	Phase	Criterion	Result
erc	P4	errors + warnings = 0	pass 0/0
place	P6	legality, edges, decouplers	pass 0/0
drc_routed	P7	parity + all track checks, err+warn = 0	pass 0/0
verify	P8	8-check suite, error severity = 0	pass 8/8
dfm	P9	gerber-side DFM, error severity = 0	pass 0 err

Two independent adversarial reviews (schematic, board) ran with fresh context. Notable outcomes: the schematic review verified the window truth table at 4.4/5.25/6.7/9/20 V and the transistor pin map against the manufacturer table; the board review caught (and the pipeline fixed) an initially undersized ground-return attachment at J1 and the then-missing functional silk package.

6.1 Standing waivers

- **W1** DP/DM short-at-chip: accepted as hardware-safer; PD-only capability is the documented cost.
- **W2** CH224K quiescent current unpublished: bounded by arithmetic (0.979 mA worst corner vs. 1.7 mA budget) and by WCH's own reference circuit; measured at bring-up.
- C1B bulk-cap feed crosses the pour through 1.11+1.25 mm channels (checker-blind geometry, disclosed): dead-end stub, ~0.3 A ripple vs. 3.49 A capacity.
- J1 mouth sits ~0.5 mm inboard of the board edge; thick cable overmolds may not latch — respin note.

7 Bill of materials

24 lines, 28 placed references. Ext. = JLC Extended (feeder fee applies). Prices are qty-1 LCSC references.

MPN	Value / description	Package	LCSC	Tier	\$
CH224K		ESSOP-10-150mil-1mm	C970725	Ext.	0.389
USB4105-GF-A-120	USB-C receptacle 16P SMT	SMD	C5184243	Ext.	0.9247
KF128-5.08-2P-AA	Screw terminal 5.08mm 2P	Plugin,P=5.08mm	C474952	Ext.	0.19
HX PZ2.54-1x2P ZZ	1x2 2.54mm male THT	Plugin,P=2.54mm	C32713268	Ext.	0.0158
TVS2200DRVR	22V flat-clamp TVS, unidirectional	WSO-6-EP(2x2)	C523793	Ext.	0.3777
BZX84C6V2	6.2V zener	SOT-23	C841160	Ext.	0.0196
XL-1608SYGC-06	Yellow-green LED 0603, Vf 2.2V	0603	C965805	Ext.	0.0074
F.0603.00011/P2-0603R4TS2-06T-001	Red LED 0603, Vf 2.2V, 100mcd@5mA	0603	C7496813	Ext.	0.0108
BC847BS	Dual NPN, SOT-363	SOT-363	C41375126	Ext.	0.0283
2.54-3P TPGT	3-position SPST DIP switch, SMD 2.	SMD,P=2.54mm	C7421520	Ext.	0.4083
BSMD1206-100-30V	PPTC 1.0A hold / 1.8A trip, 30V, 1	1206	C5358568	Ext.	0.0988
CL31A106KBHNNNE	10uF 50V X5R (2x, forms the C1 22u	1206	C13585	Basic	0.3735
CC0603KRX7R9BB104	100nF 50V X7R	0603	C14663	Basic	0.0241
CL10A105KB8NNNC	1uF 50V X5R	0603	C15849	Basic	0.0966
0603WAF1002T5E	10k	0603	C25804	Basic	0.0077

MPN	Value / description	Package	LCSC	Tier	\$
0603WAF1003T5E	100k	0603	C25803	Basic	0.0079
1206W4J0511T5E	510R (x2 in series, ~1k drop-per)	1206	C25386	Ext.	0.0112
0603WAF6801T5E	6.8k	0603	C23212	Basic	0.0029
0603WAF4701T5E	4.7k	0603	C23162	Basic	0.0117
0603WAF4702T5E	47k	0603	C25819	Basic	0.0112
RC1206FR-073K3L	3.3k	1206	C137292	Ext.	0.0178
0603WAF1501T5E	1.5k	0603	C22843	Basic	0.0023
0805W8F4701T5E	4.7k	0805	C17673	Basic	0.011
0603WAF0000T5E	0R	0603	C21189	Basic	0.0027

8 Ordering and bring-up

Estimated \$19.65 for 10 assembled (\$1.96/board; the JLC cart is the only real quote). Human steps before ordering, from `fab/order.json`:

1. **Select 2 oz copper** on the JLC order page (the 5 A path depends on it; the default is 1 oz).
2. Upload the gerber zip to `jlcdm.com` and the quote page; review their DFM report.
3. In the assembly preview, check every polarized part's rendered orientation (LEDs, zener, TVS, tantalum-free board — diode-class parts).
4. Hand-solder J2/J3 through-hole parts after SMT delivery.
5. Bring-up: measure /VDD at the 5 V profile (W2); buzz out Q1 E/B/C once; continuity-check SW1 pins 1–3 unpressed; use an e-marked cable and a genuine PD source for any >3 A draw.